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Machine Learning Assignment #5

5a.) Range Normalization (0,1)

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

$$X_{min} = 27$$

$$X_{max} = 89$$

$$X = \text{orig. value}$$

$$(-1,1) \cdot X' = 2 \cdot \frac{X - \min}{\max - \min} - 1$$

ID	normalization (0,1)
1	0.2419
2	0.3226
3	0.5161
4	0 ← num is 0
5	0.9194
6	0.3548
7	0.7258
8	0.2581
9	0.7419
10	0.5161
11	0.5000
12	0.8871
13	0.3710
14	0.8387
15	1
16	0.7741
17	0.6935
18	0.5161
19	0.6452
20	0.1290

ID	(-1,1)
1	-0.5161
2	-0.3548
3	0.0322
4	-1
5	0.8387
6	-0.2903
7	0.4516
8	0.5161
9	0.4839
10	0.0323
11	0
12	0.7742
13	-0.2581
14	0.6774
15	1
16	0.5484
17	0.3871
18	0.0323
19	0.2903
20	-0.7419

5c.) Mean = 60.95

-18.95 -13.95 -1.95 -33.95 23.05
 -11.95 11.05 -17.95 12.05 -1.95
 -2.95 21.05 -10.95 18.95 28.5
 14.05 9.05 -1.95 6.05 -25.95

359.1025 194.6025 3.8025 1,152.6025 531.3025

Standard deviation ≈ 16.82

$$Z = \frac{X - 60.95}{16.82}$$

ID	Standardization
1	$\approx +1.12$
2	≈ -0.83
3	≈ -0.12
4	≈ -2.02
5	≈ 1.37
6	≈ -0.71
7	≈ 0.66
8	≈ -1.07
9	≈ 0.72
10	≈ -0.12
11	≈ -0.18
12	≈ 1.25
13	≈ -0.65
14	≈ 1.07
15	≈ 1.67
16	≈ 0.84
17	≈ 0.54
18	≈ -0.12
19	≈ 0.36
20	≈ -1.54

8a.) Using Pandas in Python, I was able to put the dataset into the program and print the correlation (≈ -0.9607).

8b.) By studying the graph it seems like the higher the life expectancy is, the more high of a chance for there to be an Infant mortality (A negative relationship). There also seems to be multiple parabolic relationships. There is a negative one between Infant Mortality and Health USD, a negative one between Infant Mortality / Health USD.